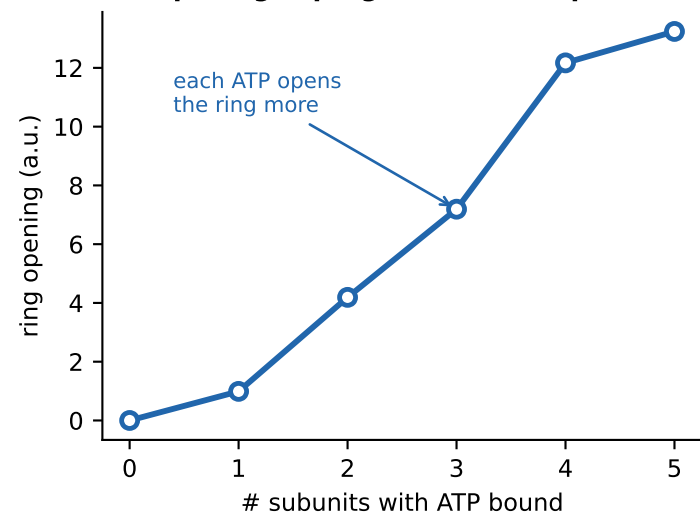
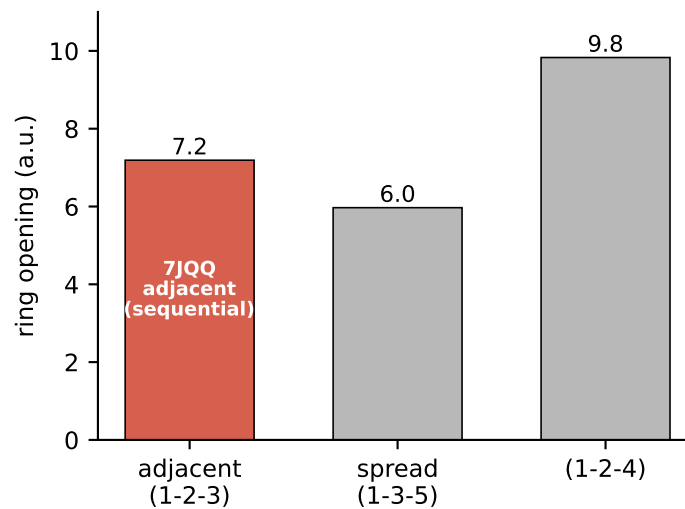


gp16 ring opening: an ATP-occupancy mechanism that structure predictors miss

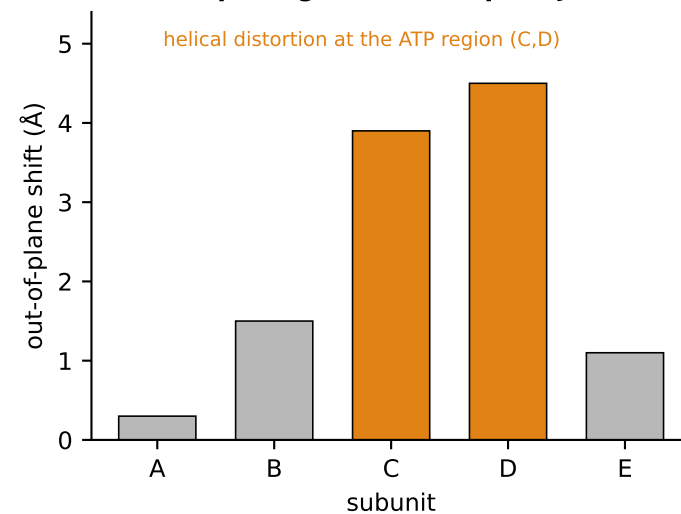
A Opening is progressive & cooperative



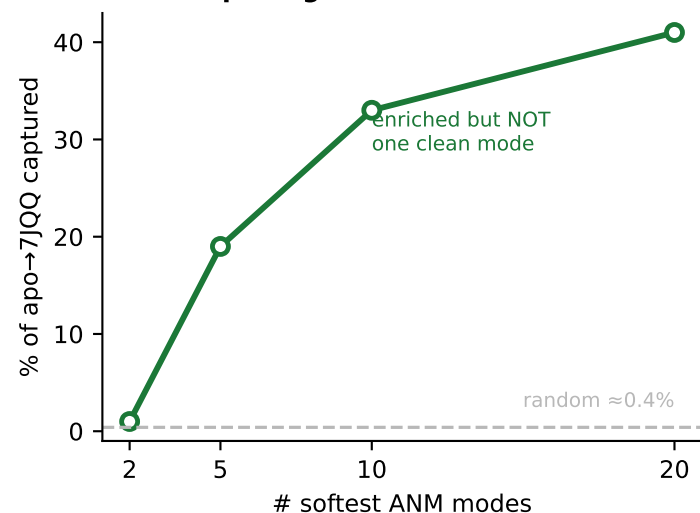
B 3-ATP: which subunits sets the state



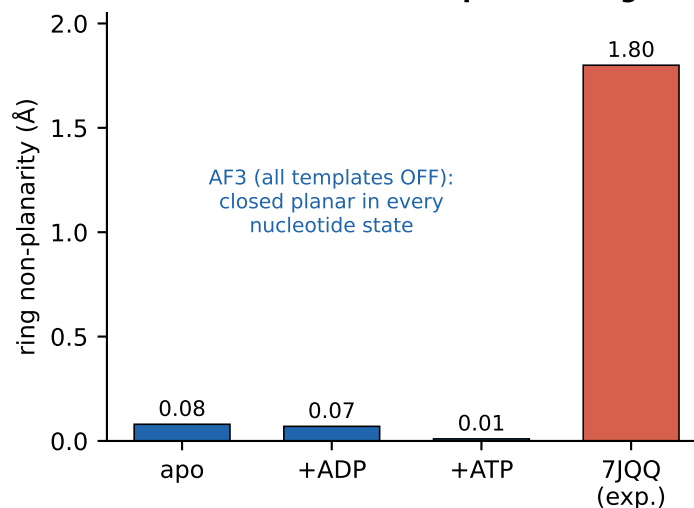
C Opening localizes (apo→7JQQ)



D Opening is soft-mode-enriched



E Predictors can't open the ring



The mechanism prediction

- ATP opens the ring progressively & cooperatively (A), and the OPENING DEPENDS ON WHICH subunits fire (B).
 - Sequential firing → an ADJACENT ATP block (7JQQ pattern, opening 7.2); concerted/spread → 6.0 or 9.8.
 - Static predictors give a closed planar ring for apo/ADP/ATP alike (E) — the opening is real physics they miss (D).
- A covalent addressable ring + single-molecule tweezers can SET the ATP pattern and read step/force → decide sequential vs concerted.